Government of Canada Gouvernement du Canada

DAOD 6500-0, Data Management and Analytics

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1. Introduction

Date of issue: 2021-11-05

Application: This DAOD is a directive that applies to employees of the Department of National Defence (DND employees) and an order that applies to officers and non-commissioned members of the Canadian Armed

Forces (CAF members).

Approval Authority: Assistant Deputy Minister (Data, Innovation and Analytics) (ADM(DIA))

Enquiries: Director Data Policy and Digital Innovation (DDPDI)

Definitions

analytics (*analytique*)

Systematic computational transformation of data into insights, for the purpose of making better decisions. (Defence Terminology Bank record number 696416)

artificial intelligence (*intelligence artificielle*)

Information technology that performs tasks that would ordinarily require biological brainpower to accomplish, such as making sense of spoken language, learning behaviours, or solving problems. (*Directive on Automated Decision-Making*, Treasury Board)

big data (*mégadonnées*)

Data produced in high volume, at high speed and in various formats, and which is too complex to be handled with traditional data-processing software. (Defence Terminology Bank record number 695863)

business intelligence (*intelligence d'affaires*)

The applications, infrastructure, tools and best practices that enable access to, and analysis of, information to improve and optimize decisions and performance. (Defence Terminology Bank record number 695860)

data (*données*)

Set of values of subjects with respect to qualitative or quantitative variables representing facts, statistics, or items of information in a formalized manner suitable for communication, reinterpretation, or processing. (*Policy on Service and Digital*, Treasury Board)

Note – In the DND and the CAF, data is created, collected and used both in military operations and exercises, and in corporate administrative processes.

data architecture (architecture des données)

An integrated collection of master design documents at different levels of abstraction that govern how data is collected, stored, arranged, used and removed.

Note – Data architecture is classified by descriptions of all the containers and paths that data takes through an organization's systems. (Defence Terminology Bank record number 695861)

data asset (*ressource des données*)

An entity comprised of data from any source that can be governed and managed and that has potential to provide value or produce benefit. This can include data sets, databases, big data, and system and application output files. (Defence Terminology Bank record number 696417)

data governance (*gouvernance des données*)

A system of decision rights and accountabilities applicable to data-related processes.

Note – This system describes which action can be taken on which data asset, by whom, under which circumstances and using which methods. (Defence Terminology Bank record number 695865)

data integration (*intégration des données*)

The movement and consolidation of data within and between data stores, applications and organizations, into consistent forms either physical or virtual. (Defence Terminology Bank record number 695866)

data management (*gestion des données*)

The development, execution and supervision of plans, policies, programs and practices that deliver, control, protect and enhance the value of data and information assets throughout their life cycles. (Defence Terminology Bank record number 27521)

data quality (*qualité des données*)

A degree or level of confidence that the data provided meets requirements of the data user in terms of characteristics such as accuracy, completeness and reliability. (Defence Terminology Bank record number 33436)

data steward (*responsable des normes de données*)

The individual responsible for the life cycle of the data in a specific system or functional area. (Defence Terminology Bank record number 33440)

digital (*numérique*)

Processes, practices and technologies related to the production, storage, processing, dissemination and exchange of electronic information and data. It refers to, among other things, information and communications technologies, infrastructures, and the information and data they produce and collect. (*Policy on Service and Digital*, Treasury Board)

domain (*domaine*)

A specific field of knowledge or expertise. (Defence Terminology Bank record number 21857)

enterprise data model (*modèle de données d'entreprise*)

A holistic, enterprise-level, implementation-independent conceptual or logical data model that provides a common and consistent view of data across the enterprise. (Defence Terminology Bank Record number 696418)

information (*information*)

The representation that a human or a machine assigns to data, facts or knowledge by means of known conventions such as reports, events, processes, decisions, ideas or opinions in any medium or form. (Defence Terminology Bank record number 696374)

information asset (*ressource d'information*)

A body of information defined and managed as a single unit so the body of information can be understood, shared, protected and exploited effectively.

Note – An information asset has recognizable and manageable value, risk and content for DND business and CAF operations. (Defence Terminology Bank record number 696375)

interoperability (*interopérabilité*)

The ability of different types of electronic devices, networks, operating systems, and applications to work together effectively, without prior communication, to exchange information in a useful and meaningful manner. (*Policy on Service and Digital*, Treasury Board)

master data (*données maîtres*)

Data that provides the context for business activity data in the form of common and abstract concepts that relate to the activity.

Note – Master data includes the details (definitions and identifiers) of internal and external objects involved in business transactions such as customers, products, employees and vendors. (Defence Terminology Bank record number 695867)

metadata (*métadonnées*)

Data that describes data itself, the concepts the data represents and the connections between the data and concepts. (Defence Terminology Bank record number 695869)

predictive analytics (*analyse prédictive*)

The use of regression analysis, forecasting, multivariate statistics, pattern matching, predictive modeling or other techniques to provide insight as to what could happen in the future. (Defence Terminology Bank record number 696419)

reference data (*données de référence*)

Data that defines a set of permissible values that are used by other data fields and do not generally change, for example, units of measurement and country codes. (Defence Terminology Bank Record number 696420)

3. Policy Direction

Context

3.1 The modern defence environment requires that data be leveraged in all aspects of defence programs, at or near-real time, enhancing defence capabilities and speed and effectiveness in decision-making, and providing an information advantage during military operations.

3.2 The DND and the CAF must therefore establish, govern and operationalize a defence data management function that provides robust governance of DND and CAF data and enables high-quality data to be leveraged as a strategic asset.

3.3 The DND and the CAF must utilize their data to support the capability for analytics at all levels.

3.4 The *CDS/DM Joint Directive on Data Management* provides that ADM(DIA) is the functional authority for:

a. data management, data governance, analytics and exploitation of advanced analytics and artificial intelligence; and
b. the development and operationalization of the data and analytics program capability.

Purpose

3.5 This DAOD provides the strategic direction for DND and CAF policies, directives, instructions, standards and guidelines relating to data management and analytics.

Policy Statement

3.6 The DND and the CAF are committed to using and promoting data and analytics as strategic assets by:

a. enabling authorized DND employees and CAF members to access, collect and manage data in all aspects of DND and CAF defence programs to enhance defence capabilities and reporting;
b. enabling access and control of data across the DND and the CAF to provide insight, operational awareness, trust and effectiveness;
c. enabling the optimisation of data for self-serve, near real-time analytics, visualisation, digital services and decision-making; and

d. ensuring that data can be used for business intelligence, predictive analytics and artificial intelligence.

Requirements

3.7 The DND and the CAF must:

a. create a culture whereby data is leveraged in all decisions, and all DND employees and CAF members are held accountable for their role in data management;
b. support a data governance framework in which the roles and responsibilities of data stewards are delineated, monitored and audited;
c. support a flexible risk-based security approach that balances the provision of greater internal access to data with the maintenance of data-centred security, ethics and protection of intellectual property rights;
d. ensure that the privacy of individuals is protected in accordance with the Privacy Act, including in the context of data interoperability;
e. develop and maintain an enterprise data model and related data glossaries and dictionaries to support decision-making, task effectiveness and efficiency of work, based on systems of record and authoritative data sources;
f. align with Government of Canada policies, directives and practices on data governance, stewardship, enterprise and data architecture, and the management of master data, reference data, metadata and data quality;
g. maintain data quality by entering accurate data into the appropriate systems of record in a timely manner and taking corrective action to address data integrity issues;

h. ensure that data is managed to reduce redundancy and enable interoperability with the North Atlantic Treaty Organization, allies and other government departments;
i. ensure that data are stored in appropriately accredited repositories, commensurate with the security designations and markings of the relevant data;
j. ensure that data are guarded against unauthorized access, disclosure or destruction;
k. enable releasable, open data to be added to the Government of Canada's Open Government portal, informed by public demand;
l. ensure that data is managed in accordance with data retention and disposition directives, standards and guidelines; and
m. support DND employees and CAF members in becoming data literate.

4. Consequences

General

4.1 Should clarification of the policy statement, requirements or authorities be required, DND employees and CAF members may seek direction through their channel of communication or chain of command, as appropriate. The anticipated results set out in the policy statement may not be achieved by the DND and the CAF if the requirements specified in this DAOD are not properly implemented. Not achieving the anticipated results could affect the ability of the DND and the CAF to ensure that the CAF is prepared to undertake missions for the protection of Canada and Canadians and the maintenance of international peace and stability.

5. Authorities

5.1 The following table identifies the authorities associated with this DAOD:

The ...	has or have the authority to...
Assistant Deputy Minister (Information Management)	• ensure that appropriate information technology security measures are applied to all DND and CAF data, activities and processes, in accordance with DAOD 6003-0, *Information Technology Security*; • oversee the effective and efficient management of systems and technologies and their related databases; • oversee the release of DND and CAF data as an open resource to the Government of Canada Open Government Portal; and • negotiate required changes to the retention and disposition authorities and schedules with Library and Archives Canada.
Chief of Defence Intelligence (CDI)	• exercise functional authority with respect to data collected under all DND and CAF defence intelligence programs, administration and activities, including deployed operations.

ADM(DIA)	• establish strategic direction and governance structure for data management; • issue frameworks, policies, directives, standards and guidelines on data management (data architecture, data modelling, master data management, reference data management and data quality management), analytics and artificial intelligence, in consultation with data stewards, data governance partners and data consumers, and oversee compliance with these instruments; • approve and oversee plans and programs pertaining to data management; • oversee the DND and the CAF data architecture and data models for alignment with Government of Canada, DND and CAF policies, directives, standards and guidelines; • expand and mature an analytics capability with the potential to inform every decision maker; • promote data-enabled business transformation; • support the use of tools and training to assist practitioners in both their understanding of data management and their easy and consistent delivery of high quality and curated data, as part of their normal operating model; • support opportunities to innovate using data, analytics and data-enabled disruptive technologies; and • establish strategic direction on data literacy and a culture of evidence-based decision-making.

level one advisors who are identified as data domain stewards and CDI	• issue amplifying directives, instructions and standards, other than DAOD, on data management and analytics; • ensure timely and accurate data entry; • collaborate with ADM(DIA) on best practices on data collection, management, use, dissemination, analysis and reporting; and • maintain the overall integrity of the data for which they are the recognized data domain steward.
Corporate Secretary	• advise and provide policy guidance on data governance for purposes of the administration of the *Privacy Act*.

6. References

Acts, Regulations, Central Agency Policies and Policy DAOD

- *Access to Information Act*
- *Financial Administration Act*
- *Library and Archives of Canada Act*
- *National Defence Act*
- *Official Languages Act*
- *Privacy Act*
- *Policy on Service and Digital*, Treasury Board
- *Directive on Automated Decision-Making*, Treasury Board

- *Directive on Service and Digital*, Treasury Board
- *Guideline on Service and Digital*, Treasury Board
- DAOD 1000-6, *Policy Framework for Information and Information Technology*
- DAOD 1000-8, *Policy Framework for Safety and Security Management*

Other References

- DAOD 1001-0, *Access to Information*
- DAOD 1002-0, *Administration of the Privacy Act*
- DAOD 2011-0, *Enterprise Architecture*
- DAOD 3000-0, *Materiel Acquisition and Support*
- DAOD 3008-0, *Intellectual Property*
- DAOD 6000-0, *Information Management and Information Technology*
- DAOD 6001-0, *Information Management*
- DAOD 6001-1, *Recordkeeping*
- DAOD 6003-0, *Information Technology Security*
- DAOD 7023-0, *Defence Ethics*
- DAOD 8008-0, *Defence Intelligence*
- *CDS/DM Joint Directive on Data Management* (Accessible only on the National Defence network.)
- *Values and Ethics Code for the Public Sector*
- *Department of National Defence and Canadian Forces Code of Values and Ethics*
- *Accountabilities, Responsibilities and Authorities Framework for ADM(DIA)* (Accessible only on the National Defence network.)
- *The Department of National Defence and Canadian Armed Forces Data Strategy*
- *Data Governance Framework*

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